**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 055 · Selection: IF Statements

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When is A AND B true?

Q2. What does CPU cache hold?
A. Only saved personal files
B. Only website passwords
C. Often-used data and instructions
D. Only the next address

Q3. “A register is a large external drive used for backups.” Correct this. Explain why.

Q4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Selection — Selection lets a program follow different instructions according to a condition.

IF — An IF block executes its instructions only when the condition is true.

Comparison — The Boolean test must compare values rather than assign a new value.

ENDIF — ENDIF marks the end of the IF statement so later instructions can run normally.

Algorithm — An ordered set of instructions to solve a problem.

Trace — Following instructions and recording changes to values or output.

## New learning

What is Selection?: Selection — Selection lets a program follow different instructions according to a condition.

In this example: A theme-park ride height rule uses if; An IF block executes its instructions only when the condition is true.

Follow the effect: The Boolean test must compare values rather than assign a new value. Explain how this changes the eventual program result.

Check understanding: ENDIF is used for a specific reason: ENDIF marks the end of the IF statement so later instructions can run normally.

## Worked example

A theme-park ride height rule is being written. Explain how selection and if are used to make the program work. [4 marks]

1. We are designing a theme-park ride height rule. Identify which data and decisions are essential.

2. Selection [1] — Selection lets a program follow different instructions according to a condition. [1].

3. IF [1] — An IF block executes its instructions only when the condition is true. [1].

4. Two linked points for each technique give four mark points.

# We Do · Practise with guidance

### Guided 1

In a school competition entry, explain how selection helps the program. [2 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: Selection [1] — Selection lets a program follow different instructions according to a condition. [1].

### Guided 2

For a sports pass check, explain why if matters. [2 marks]

Worked method: IF [1] — An IF block executes its instructions only when the condition is true. [1].

### Guided 3

When designing a exam pass notification, explain how comparison and endif should be used. [4 marks]

Worked method: Comparison [1] — The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] — ENDIF marks the end of the IF statement so later instructions can run normally. [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Complete the IF condition that switches on ventilation above 30 degrees.

```
temperature = 32
if ... then
    print("Fan on")
endif
```

write

### Question 1

State two programming terms or techniques relevant to this lesson. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 2

A cinema age check needs to work correctly. Explain how if and comparison would help. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

### Question 3

A pupil is programming a library opening status. Explain the roles of selection, comparison and endif in their solution. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Exit Ticket · Answer before checking

Selection lets a program follow different instructions according to a condition.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Only when both A and B are true.

Q2. C — Often-used data and instructions

Q3. A small, fast storage location inside the CPU. It holds data or addresses in use.

Q4. The accumulator holds the result, 18. The PC holds the next instruction's address, 240.

### I Do / We Do pages · Guided practice

Q1. Selection [1] — Selection lets a program follow different instructions according to a condition. [1].

Q2. IF [1] — An IF block executes its instructions only when the condition is true. [1].

Q3. Comparison [1] — The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] — ENDIF marks the end of the IF statement so later instructions can run normally. [1].

### You Do pages · Code First

```
temperature = 32
if temperature > 30 then
    print("Fan on")
endif
```

Expected result: Fan on

### You Do pages · Independent questions

Q1. Selection [1]; Comparison [1].

Q2. IF [1] — An IF block executes its instructions only when the condition is true. [1]. Comparison [1] — The Boolean test must compare values rather than assign a new value. [1].

Q3. Selection [1] — Selection lets a program follow different instructions according to a condition. [1]. Comparison [1] — The Boolean test must compare values rather than assign a new value. [1]. ENDIF [1] — ENDIF marks the end of the IF statement

so later instructions can run normally. [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**